BATCH LEARNING

- The conventional machine learning is rooted in the *statistical learning theory* and is sometimes referred to as the *batch learning scenario*:
  - A data set $\mathcal{D} = \{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$ is given beforehand in form of a random sample (iid observations).

Advanced Machine Learning

Introduction to Online Learning

Learning goals

- Understand the difference between batch and online learning
- Know the basic and the extended learning protocol in online learning
- Know how performance is measured in online learning

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  - *Sequential order* — data is generated only bit by bit;
  - *On-the-fly decisions* — decisions or predictions have to be made during the data generating process;
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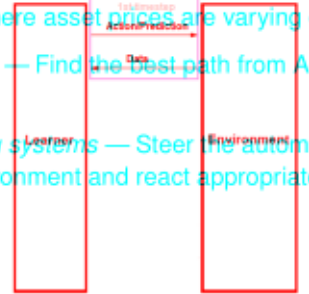

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THE BASIC ONLINE LEARNING PROTOCOL

Formally, an online learning problem consists of

- a learner (forecaster, agent resp. decision maker), an environment (user resp. adversary, system resp. nature),
- time steps $1, 2, \ldots, T$ (may be infinite),
- available actions $\mathcal{A}$ for the learner (may be infinite),
- environmental data space $\mathcal{Z}$,
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- the learner's chosen action $a_t = \hat{y}_t$ corresponds to a prediction,
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- In particular, the environmental data are not necessarily generated by a probability distribution!
- The protocol of the online learning problem is now as follows. In each time step t
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- In particular, the environmental data are not necessarily generated by a probability distribution!
- This also covers the area of *adversarial learning*: the data can even be generated by an adversary trying to fool the learner.
- However, the online learning setting can of course also be considered in a statistical setting.

ONLINE LEARNING: REQUIREMENTS

- The dynamical aspects have to be incorporated for the design of efficient learning algorithms.
- The online learner has to cope with the sequential availability of the data and to cope with time as well as computational constraints.
- Roughly speaking, one seeks to construct an online learning algorithm which is adaptive to the environment and allows incremental as well as preferably cheap updates over time.

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- Although consideration of time and memory constraints is important for practical purposes, we will only implicitly consider these constraints in this lecture.
- We will mainly focus our theoretical analysis on the performance of the learner in terms of its (cumulative) loss, which, however, will usually ignore computational aspects of the learner.

MEASURE OF QUALITY IN ONLINE LEARNING

- In order to measure the quality of an online learner one can compute the difference between the cumulative loss of the learner and the cumulative loss by taking some competing action $a \in \mathcal{A}$:

$$R_T(a) = \sum_{t=1}^{T} L(a_t, z_t) - \sum_{t=1}^{T} L(a, z_t).$$

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- Formally, the following should hold

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Interpretation: The average regret per time step (or per example) goes to zero:

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DYNAMIC REGRET

- One might ask why one compares only with a fixed best action in hindsight, say a^*, instead of a sequence of actions $a_1^*, a_2^*, \ldots, a_T^*$?
- The rationale behind this measure of quality is that the best fixed action in hindsight is already reasonably good over all the time steps: it performs almost as well as a batch learner that observes the entire sequence and picks the best action in hindsight.
- However, this is too optimistic and may not hold in changing environments, where data are evolving and the optimal action is drifting over the time.
- To address this limitation, recent works have also considered the *dynamic regret*:

$$R_T^D(a_1^*, a_2^*, \ldots, a_T^*) = \sum_{t=1}^{T} L(a_t, z_t) - \sum_{t=1}^{T} L(a_t^*, z_t).$$

- We will cover only the static regret in this lecture.